Predicting Urban Farming Participation Intention in the National Capital Region, Philippines using Machine Learning Approaches

Aviona Bianca S. Bernil
College of Computer Studies and Multimedia Arts, FEU Institute of Technology
Manila, Philippines
asbernil@fit.edu.ph

Jericho Nheil C. Colina
College of Computer Studies and Multimedia Arts, FEU Institute of Technology
Manila, Philippines
jccolina@fit.edu.ph

Godwin Aldrich M. Ablao
College of Computer Studies and Multimedia Arts, FEU Institute of Technology
Manila, Philippines
gmablao@fit.edu.ph

Alexander A. Hernandez
College of Computer Studies and Multimedia Arts, FEU Institute of Technology
Manila, Philippines
aahernandez@feutech.edu.ph
*Corresponding Author

Roman M. De Angel
College of Computer Studies and Multimedia Arts, FEU Institute of Technology
Manila, Philippines
rmdeangel@feutech.edu.ph

***Abstract*—This study employs predictive analytics through supervised machine learning to examine how environmental knowledge, green innovation, and AI-technological sustainability influence urban farming participation intention in the city of Manila. Quantitative research was conducted, through an online survey from the city of Manila residents, irrespective of prior farming experience. Six machine learning models, random forest, k-nearest neighbors, support vector machine, decision tree, logistic regression and gradient boosting, were trained and tested after the data preprocessing to determine the strongest predictors of urban farming participation intention. The Random Forest model, thus, found to be the most accurate in predicting, demonstrated best R^2, with the lowest error rates for predicting urban farming intention. Both the feature importance and SHAP analyses identify environmental knowledge as the primary driver, followed by AI technological sustainability and green innovation. The results show that people with higher environmental literacy and who have better access to AI-driven technologies are more likely to participate in urban farming. The study concluded that the city of Manila urban farmers would benefit through environmental education technological accessibility, and green innovation. Research and practical implications are presented to extend the contribution of the study.**

Keywords—urban farming, green innovation, environmental knowledge, analytics modelling, machine learning, artificial intelligence, adoption intention, technology sustainability.

I. Introduction

Urban farming is being promoted more as a sustainability strategy in highly urbanized areas such as Metro Manila, where food security and environmental resilience are identified as the hot issues of local policies [1], [10]. The recent initiatives mainly focus on the use of digital tools, smart farming technologies, and innovation frameworks as the means to the end of community-based agriculture [2], [5], which is in line with the sustainable development goals regarding innovation, climate action, and urban sustainability [6]. Along with this, a study published recently also points out that AI can become the major technological driver behind sustainable urban agriculture. Chowdhury, Paul, and Ahmed point out that AI-led systems are instrumental in achieving less resource use, automation, and environmental friendliness for vertical farming systems, thereby putting emphasis on the role of AI in securing agricultural resilience over time.

However, the rising number of articles on the issue also reflects agreement with these ideas that environmental awareness, green innovation mindset, and AI technology access and sustainability have become the most significant factors that persuade people to engage in urban farming activities rather than given access to physical resources [3], [4], [21]. Participation level in urban farming, however, is still not sufficient in some districts of Metro Manila due to the lack of AI-related technological resources in certain communities and the different stages of innovation readiness among them [8], [9].

Urban farming have been measured through several research projects. However, the studies mainly deal with technical areas and do not provide predictive modeling for identifying which factors most influence participation [3], [7]. Besides, there are very few empirical studies that rank environmental knowledge, AI technology sustainability, and innovation attitude as the most important leading factors of actual participation, especially in the context of highly urbanized Filipino communities [2], [9], and other similarly situated countries.

This research is aims at bridging the gap by producing a prediction model of participation in urban farming with environmental knowledge, green innovation, and AI technology sustainability as independent variables. This study will use a supervised machine learning method with a Random Forest Regressor, K-Nearest Neighbors (KNN), Support Vector Machine (SVM). Decision tree, Logistic Regression, and Gradient Boosting to analyze the responses of the surveys conducted among urban farming communities in Metro Manila. This study carries out descriptive analysis, reliability

validation, and feature importance extraction as part of the process [4], [5]. Findings from this study provide support to Local Government Unit (LGU) sustainability networks and policy frameworks through the generation of an evidence-based ranking of predictors that can efficiently guide the targeted design of AI-integrated urban farming interventions [1], [6], [9].

II. RELATED LITERATURE

A. Urban Farming

Urban farming is viewed as a response to urbanization and food insecurity in several developing cities and among others in Southeast Asia [1], [3]. It has been emphasized in recent literature that sustainable urban agriculture improves food supply chain systems and, furthermore, contributes to better environmental sustainability and social inclusiveness [1], [10]. Marchand and De Cian [1] posit urban farming as social innovation model that promotes community collaboration and local sustainability, while Dizon and Alvarez [2] indicated that the management of technology systems would help the effectiveness and scalability of urban farming in the metropolitan areas.

In addition, the literature also informs us that environmental knowledge, green technology innovation, and AI technological accessibility to technological systems are important determinants of participation in urban farming [3], [4], [6], [22]. Ahmad and Salleh [3] showed that environmental knowledge and attitudes are seen as an important factor in eliciting local environmental awareness of the urban agriculture. Likewise, Fei et al. [6] and Areola and Dacones [10] indicate that innovation and integration with local policies tend to be the important determinants of sustainability in cities that are based on farming systems. Bourgeois et al. [5] and Muller and Rossi [4] also express agreement with the postulation that the successfulness of the digital systems of agriculture equates with the preparedness of farmers to adopt technology for sustainability. When farmers use AI tools like precision farming and automation, digital systems work effectively [24], [25].

However, participation levels remain inconsistent across urban communities, largely due to disparities in AI technological accessibility gaps such as internet, digital literacy, affordability and innovation adoption [8], [9], [23]. Empirical studies in the Philippines are limited, as most investigations focus on rural regions or descriptive evaluations of community farming programs [7], [9]. Gao [11] and Billah [15] pointed out that, to foster urban agriculture successfully, it is necessary to have both technical and social awareness, thus making green innovation a common goal for environmental and communal wellbeing. AI promotes sustainability, reduces labor costs, and boosts crop efficiency [25].

B. Machine Learning and Analytics in Urban Farming

The application of analytics and machine learning in Urban Farming; agriculture has expanded rapidly and provides predictive information with respect to sustainability, yield prediction and participatory modeling [12], [16]. Fei [12] established that the efficiency of urban and peri- urban agriculture can be enhanced through technological innovation and data-centric systems as a result of sensor networks and IoT applications. Richter [13] demonstrated that data-driven systematic approaches to analysis can identify behavioral motivators and constraints in community driven smart farming programs.

Further, Yuan [14] offered a comprehensive overview of the integration of the socio-economic factors interacting with technological aspects of predictive models of agricultural production, when Bourgeois et al. [5], investigated the adoption of smart farming networks and their digital revolution through data analysis. Machine learning algorithms such as Random Forests, Support Vector Machines (SVM) and Artificial Neural Networks (ANN) have been deployed in predicting participation and technology acceptance patterns in agricultural systems [16].

More recently, Yuniarsih et al. [16] Botero-Valencia et al. [17], and Ali et al. [18] have presented research that demonstrated the efficacy of Random Forest algorithms in recognizing significant determinants of farming behavior, including environmental awareness and technology readiness. These observations confirm the intention of the present study to apply a supervised learning algorithm in predicting participation in urban farming based on key environmental and technological factors. Furthermore, data analytics used in sustainability programs assists local government decision making and promote evidenced based policy making in urban agriculture [19, 20, 21].

C. Conceptual Framework

Figure 1. in environmental knowledge, green innovation, and AI-technological accessibility are the independent variables that have a direct effect on urban farming, the dependent variable. It shows the hypothesized relationships, indicating that higher environmental awareness, innovation and AI technological sustainability may lead to higher participation in urban farming.

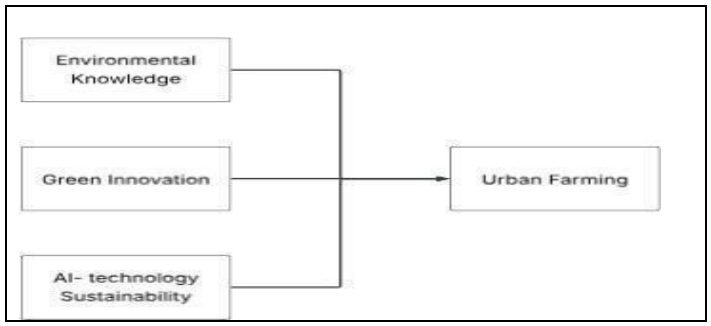

Fig. 1. Conceptual Framework

Data was gathered through an online survey directed at the residents of Metro Manila. The variables include Environmental Knowledge (EK), Green Innovation (GI), AI-Technology Sustainability, and Urban Farming Participation Intention (UFPI).

III. MATERIALS AND METHODS

A. Survey Development

Urban farming participation is measured by developing a survey instrument, based on past studies [4, 13]. The survey instrument was divided into two sections, that captures demographic profile and factors on urban farming participation such as environmental knowledge (4 items), green innovation (6 items), AI-technology sustainability (6 items), urban farming participation (4 items). The survey used a 5-point Likert's scale to determine agreement or disagreement for each item. The first version of the survey was initially pilot tested to determine the reliability and validity of the factors on urban farming participation intention. Technology adoption experts were consulted to review the survey instrument, and comments were addressed before the final version was created.

B. Data Collection

In this study, a structured online questionnaire survey is used to gather quantitative data from the residents of the City of Manila. This study applies area-based multi-stage cluster sampling (barangays to households) augmented with adaptive cluster sampling to improve efficiency for clustered/rare urban farmers. Before the data collection, consent form was provided to ensure that the research conforms to ethical standards, and prevailing data privacy law in the Philippines. The survey was conducted from August to November 2025, where two follow up sessions were conducted to increase the number survey responses.

TABLE I. DEMOGRAPHIC PROFILE OF THE RESPONDENTS

Attribute	Category	Frequency	Percentage
Age	20-24 years old	327	81.75%
	25-34 years old	36	9.0%
	35-44 years old	15	3.75%
	45-55 years old	17	4.25%
	55 years old above	10	1.25%
Gender	Male	195	48.8%
	Female	199	49.8%
	Prefer not to say	6	1.5%
Education	Elementary	3	0.8%
	High school	39	9.8%
	Vocational / Technical	10	2.5%
	Bachelor's	343	85.8%
	Master's / Doctorate	5	1.2%
Occupation/ Type of work	Employed(private sector)	320	80%
	Employed (government)	30	7.5%
	Self employed	17	4.2%
	Unemployed	33	8%
Monthly household income	Below 20,000	81	20.2%
	20,000 – 40,000.00	191	47.8%
	40,001 – 60,000.00	30	7.5%
	Above 60,000.00	39	9.8%
	Prefer not to say	59	14.7%

C. Data Preprocessing

The survey data was processed and is ready for the algorithms to work. Responses recorded in Google Forms were exported into a spreadsheet and changed into a CSV file, used in Python. All incomplete and improper entries were eliminated, and uniformity of variable formats resulted. Categorical responses were encoded into numerical values to allow processing by the machine learning algorithms. Those items based on the Likert scale were coded, 1 through 5, in the order of Strongly Disagree to Strongly Agree, and obtained their respective composite sums, which were the measurements needed for environmental knowledge, green Innovation, AI technological sustainability, and urban farming participation. All numerical variables were scaled to the same range to improve the performance of KNN and SVR models. Further, extreme values were identified and removed to reduce the effect of outliers on model performance.

The demographic data were encoded in a suitable manner one-hot encoding and prepared for the computational work. It was also useful to note and eliminate outlying and inconsistent data before its usages for purposes of integrity. Training and testing versions of the final product were separated. 20% and 80% of the subsets, respectively, were useful for validating the model. All preprocessing was carried out in Python employed the following libraries: Pandas, NumPy, and Scikit-learn.

D. Model Development

In this stage, six supervised machine learning algorithms which are Randon forest Regressor, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree (DT), Logistic Regression (LR), Ridge, and Gradient Boosting Algorithm (GB). These models were used to figure out the leading factors that people get involved in urban farming. The dataset after processing was split into two parts which is 80% for training and 20% for testing. EK, GI, and AI-TS were independent variables, and urban farming was the dependent variable. Hyperparameter setting was carried out for each algorithm's performance optimization. Various parameters such as the number of trees, maximum depth, learning rate and kernel type were experimented to improve the prediction accuracy and to reduce overfitting. The dataset was divided into training and testing sets, with hyperparameters adjusted using the training data only. For each model, a range of candidate hyperparameter values was evaluated, and the combination that produced the lowest prediction error on the training set was selected. The finalized models were then assessed using the independent testing set to evaluate their generalization performance. All the models were configured to increase prediction and trained by Pythons's scikit- learn library. Different visualization tools were used to meaningfully present the results, using seaborn, and matplotlib APIs in Pyhton.

TABLE II. HYPERPARAMETER SETTINGS

Algorithm(s)	Parameters
Random Forest	{'max_depth': 10 'n_estimator': 50}
KNeighbors	{'n_neighbors': 7, 'weights': 'distance'}

Algorithm(s)	Parameters
SVR	{'C': 10, 'kernel': 'linear'}
Decision Tree	{'max_depth': None, 'min_samples_leaf':10}
Ridge	{'alpha': 1.0}
Gradient Boosting	{'learning_rate': 0.01,'n_estimators': 200}

E. Performance Evaluation

The performance of the models was evaluated using regression metrics which were used to give an in-depth analysis of the models predictive accuracy. Typical regression metrics like Mean Absolute (MAE), Root Mean Square Error (RMSE) and the coefficient determination (R^2) were used to measure how close the predicted values were to the actual participation levels. Lower MAE and RMSE values indicates smaller prediction errors, R^2 value. These metrics were instrumental and understanding model performance in detail beyond the error measures. The test was used for all the evaluations to ensure the objectivity of the assessment, thus allowing the study to identify the algorithm that had the greatest predictive power for urban farming participation intention.

IV. RESULTS AND DISCUSSION

A. Descriptive Analytics

The frequency distribution graphs display the different levels of Urban Farming Participation along with the frequencies that correspond to them in the dataset. The histogram for the continuous scores shows that the largest number of values is close to 4.0 and many of them are between 3.5 and 5.0, which indicates that the overall participation rate is high. The bar plot of the discrete categories also shows this trend as level 4.0 is the most frequently occurring, followed by levels 3.0 and 5.0. Unlike that, levels 1.0 and 2.0 are very much less frequent, thus revealing a significant class imbalance and suggesting that extremely low participation is rare. In short, the graphs manifest that most of the respondents either agree or strongly agree with the statements related to urban farming participation.

The correlation heatmap (Fig. 2) depicts how strongly and in what direction the independent variables, Environmental Knowledge, Green Innovation, and AI Technological Accessibility, and the dependent variable, Urban Farming Participation are linearly related. In fact, all three predictors have strong positive correlations with participation where Environmental Knowledge correlates the most at 0.68, followed by AI Technological Accessibility at 0.63 and Green Innovation at 0.59. The findings point to the fact that a higher level of environmental awareness, a greater receptivity to green innovation, and a better provision of AI technology are all factors that go hand in hand with increased participation rates in urban farming, thus, implying that these factors not only collectively influence but also drive the respondents' engagement in urban farming.

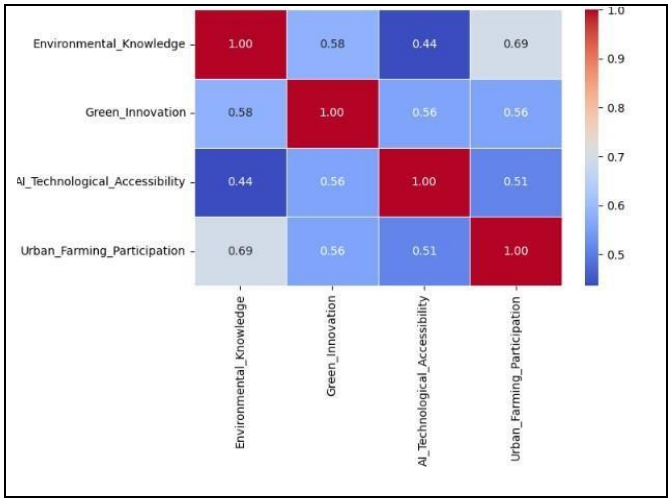

Fig. 2. Correlation Heatmap of the Urban Farming Participation Intention

B. Predictive Analytics

Table 3. provides mediate metrics for each regression model trained on the same training data, including R^2, MSE, RMSE, and MAE. The performances are higher than those on the testing set as the models have already been trained on the data. In fact, KNeighbors is remarkable with an R^2 as high as 0.8766 and very small error values, evidence that the model learns from the training data. Random Forest also produces strong results with a high R^2 of 0.8896. On the other hand, SVR ($R^2 = 0.8127$), Decision Tree ($R^2 = 0.8768$), Ridge ($R^2 = 0.8175$), and Gradient Boosting ($R^2 = 0.8417$) followed the ranks of the prediction models developed. But the presence of extraordinarily high training performances for models, especially when accompanied by significantly high scores in the testing stage, which indicates that the prediction models attained best fit.

TABLE III. MODEL PERFORMANCE (R^2)

Model(s)	Training (R^2)	Testing (R^2)
Random Forest	0.8896	0.8897
KNeighbors	0.8766	0.8656
SVR	0.8127	0.8080
Decision Tree	0.8768	0.8769
Ridge	0.8175	0.8149
Gradient Boosting	0.8417	0.8430

TABLE IV. MODEL PERFORMANCE (ERROR RATES)

Model	Test	Test	Test
	MSE	RMSE	MAE
RF	0.10	0.31	0.25
KNeighbors	0.13	0.36	0.29
SVR	0.19	0.43	0.35
DT	0.12	0.35	0.28
Ridge	0.18	0.43	0.34
GB	0.15	0.39	0.31

The feature importance analysis reveals that Environmental Knowledge (EK) is the major single factor that has the highest influence of the total factors considered,

however, there are some models which rank it lower, and that is perfectly normal (Fig. 3). Different machine learning algorithms evaluate and use features in different ways which results in natural variations of their feature importance scores. Tree- based models (e.g. RandomForest or DecisionTree) determine the ranks of features by how efficiently they reduce impurity when the data is split, therefore a different feature, such as AI Technological Accessibility sometimes slightly better at creating the first splits. Linear models (e.g. Ridge or SVR) decide importance according to coefficient sizes, thus in that case a feature with a more linear or direct relationship might become the most significant one.

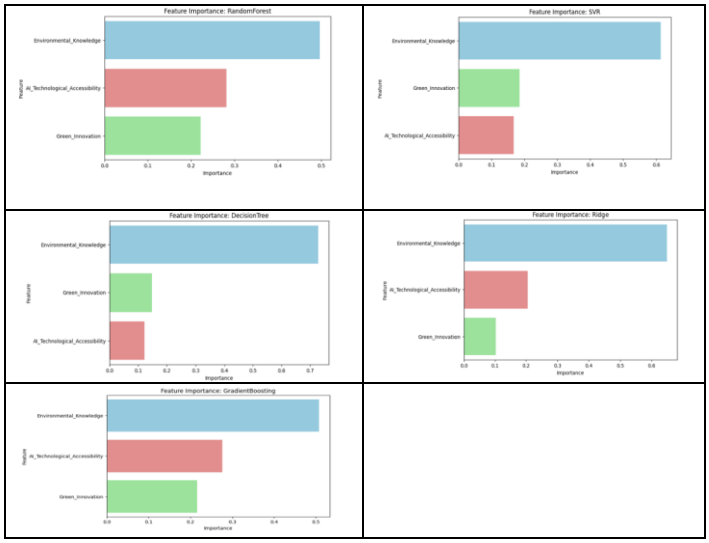

Fig. 3. Feature Importance Ranking

Non-parametric models such as KNN do not perform any calculations for importance thus it has to be obtained indirectly or will differ totally. The reasons for these differences are that every algorithm has its own unique characteristics, biases, and different ways of interpreting relationships. Some models may identify stronger linear patterns in AI Technological Accessibility or consider Green Innovation more useful for the early splits, even if Environmental Knowledge is the strongest predictor overall. Aggregated feature importance, showing the averaged over all models, is a better indicator, which lessens the individual model biases and indicates the consistent pattern when Environmental Knowledge is the main factor to predict followed by AI Technological Accessibility and Green Innovation.

The combined feature importance results show a quite clear factor hierarchy that has a major influence on Urban Farming Participation. Environmental knowledge is the main factor that is stated most clearly, which means the people who understand well the environmental problems and sustainable practices are the ones who are most likely to farm in the urban area. This fact emphasizes the importance of improving environmental literacy through specific education and awareness programs.

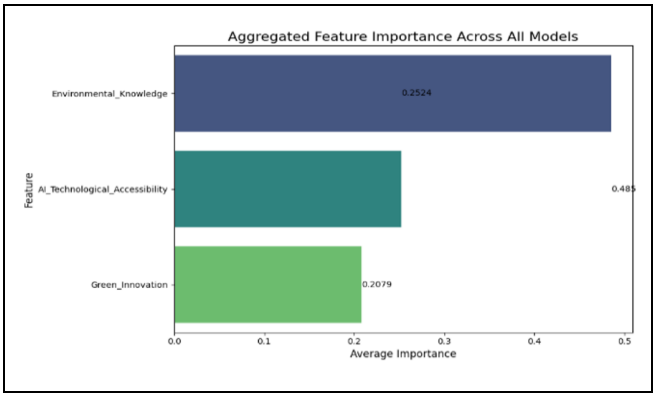

Fig. 4. Aggregate Feature Importance Ranking Across All Model

AI Technological Accessibility is the second-ranked factor, a sign that the access to smart farming tools, digital resources, and user-friendly technological support significantly promotes participation because it makes urban farming more efficient and easier to understand.Green Innovation is the third factor that indicates a situation where innovative and eco-friendly methods like hydroponics or vertical farming made a slight contribution to the participation, and they tended to come after a foundation of environmental knowledge and sufficient technological access.

C. *Discussion*

The study's findings indicate that environmental knowledge, green innovation, and AI technological sustainability are important predictors of Urban Farming Participation. Across all regression models employed in this study, the Random Forest model consistently demonstrated the strongest performance based on the highest R^2, the lowest error values. This finding suggests the relationship between our predictors and UFPI is partially nonlinear and influenced by interactions that tree-based models are particularly adept at capturing. Feature importance analysis found Environmental Knowledge was a primary driver of participation, with AI-based accessibility the next strongest added consideration for UFPI, indicating that individuals who have greater environmentally informed knowledge and access to digital or AI-based technology participation in urban farming more frequently.

Overall, the results indicate that enhancing environmental literacy, increasing access to technological resources, and stimulating green innovation can significantly promote urban farming participation in metro Manila. The prevalence of Environmental Knowledge across nearly all models suggests educational programming and awareness raising as an important starting point for producing urban farming behavior. On the other hand, the strong power of AI Technological Accessibility indicates the increasing relevance of digital tools in promoting people's ability to adopt sustainable practices. This information can assist policymakers, local government, and community organizations in developing targeted responses that combine education, innovation, and

technological assistance to promote urban farming participation.

V. CONCLUSION

The study introduces a predictive analytic framework that highlights environmental literacy, green innovation, and the sustainability of AI technology applications as critical enablers of participation in urban farming in Metro Manila. The random forest machine learning algorithm obtained the most consistent accuracy due to its capacity to suit complex, nonlinear relationships effectively. The findings suggest opportunities to participate in urban agriculture and farming through environmental education and improved accessibility of AI-related technology, while it is expected that the development of green innovations will also complement the efforts in creating sustainable innovations and practices.

Future work can build upon these findings by using longitudinal approaches to expand understanding of dynamics over time impacting urban participation, also new methods collecting real-time data using IoT, and sensor networks, may increase responsiveness in model analysis and on urban farming efficiencies. Finally, the analytics framework could be expanded by considering socio-economic factors, income and time availability), land or space constraints, household responsibilities, community support, and access to water, which are all necessary to increase the uptake of urban farming in the National Capital Region, Philippines.

REFERENCES

[1] Marchand and S. De Cian, "Urban Farming for Social Benefit: A Framework for Social Innovation in Metropolitan Regions," IFAC Papers OnLine, vol. 57, no. 4, pp. 112–118, 2024.

[2] Cacho, J. L. Technology Management Framework for Urban Farming in Selected Cities of National Capital Region for Sustainable Development. In Proc. Int. Conf. on Industrial Engineering and Operations Management (IEOM), Singapore, 2025, pp. 146–153.

[3] Yahya, A. S., Haris, N. B. M., Shah, J. A., & Zaki, N. A. (2023). Knowledge, attitude, and practice towards the adoption of urban farming: A concept paper. *International Journal of Academic Research in Business and Social Sciences*, *13*(17), 312-320.

[4] Rizzo, G., Migliore, G., Schifani, G., & Vecchio, R. (2024). Key factors influencing farmers' adoption of sustainable innovations: a systematic literature review and research agenda. *Organic Agriculture*, *14*(1), 57-84.

[5] Bekee, B., Segovia, M. S., & Valdivia, C. (2024). Adoption of smart farm networks: a translational process to inform digital agricultural technologies. *Agriculture and Human Values*, *41*(4), 1573-1590.

[6] Fei, S., Wu, R., Liu, H., Yang, F., & Wang, N. (2025). Technological innovations in urban and peri-urban agriculture: Pathways to sustainable food systems in metropolises. *Horticulturae*, *11*(2), 212.

[7] Jalotjot, H. C. (2024). Open innovation activities and capabilities of vegetable farmers in CALABARZON Region, Philippines. M.S. thesis, Graduate School of International Development, Nagoya Univ., Nagoya, Japan, 2024.

[8] Amistas, E. A. Agricultural Crop Production in the Urban Community: Its Opportunities and Challenges. Int. J. Sci. Manag. Stud., vol. 8, no. 4, pp. 222–230, 2025.

[9] Saguin, K.K. & Cagampan, M. A. Urban Farming and Land Use Governance in Metro Manila," in Governance and Sustainable Development in Southeast Asia, Springer, 2023, pp. 301–315.

[10] Areola, E. M. Q., & Dacones, D. G. (2024). Urban Farming Towards Enhancing Food Security in Highly Urbanized Cities in the Philippines, A Community Sustainability Response. *Journal of Sustainable Community Development (JSCD)*, *6*(2).

[11] Gao, P., Xiao, S., Mustapa, F. D. (2025). "A comprehensive review of urban agriculture in a changing climate: Technological innovations and policy frameworks," Climate Risk Management. c

[12] Fantini, A. (2023). Urban and peri-urban agriculture as a strategy for creating more sustainable and resilient urban food systems and facing socio-environmental emergencies. *Agroecology and Sustainable Food Systems*, *47*(1), 47-71.

[13] Richter, I., Neef, N. E., Moghayedi, A., Owoade, F. M., Kapanji-Kakoma, K., Sheena, F., & Ewon, K. (2025). Willing to be the change: Perceived drivers and barriers to participation in urban smart farming projects. *Journal of Urban Affairs*, *47*(5), 1540-1558.

[14] Yuan, G. N., Marquez, G. P. B., Deng, H., Iu, A., Fabella, M., Salonga, R. B., ... & Cartagena, J. A. (2022). A review on urban agriculture: technology, socio-economy, and policy. *Heliyon*, *8*(11).

[15] Billah, M. M., Muzahid, A., Millat, M. N., & Rahman, M. M. (2025). Urban agriculture for environmental sustainability: Perception of city dwellers. *Journal of Environmental Science, Health & Sustainability*, *1*(1), 38-50.

[16] Yuniarsih, E. T., Salam, M., Jamil, M. H., & Tenriawaru, A. N. (2024). Determinants determining the adoption of technological innovation of urban farming: Employing binary logistic regression model in examining Rogers' framework. *Journal of Open Innovation: Technology, Market, and Complexity*, *10*(2), 100307.

[17] Botero-Valencia, J., García-Pineda, V., Valencia-Arias, A., Valencia, J., Reyes-Vera, E., Mejia-Herrera, M., & Hernández-García, R. (2025). Machine learning in sustainable agriculture: systematic review and research perspectives. *Agriculture*, *15*(4), 377.

[18] Ali, Z., Muhammad, A., Lee, N., Waqar, M., & Lee, S. W. (2025). Artificial Intelligence for sustainable agriculture: a comprehensive review of AI-driven technologies in crop production. *Sustainability*, *17*(5), 2281.

[19] Weraikat, D., Šorič, K., Žagar, M., & Sokač, M. (2024). Data analytics in agriculture: Enhancing decision-making for crop yield optimization and sustainable practices. *Sustainability*, *16*(17), 7331.

[20] Rahmania, T. (2024). Exploring school environmental psychology in children and adolescents: The influence of environmental and psychosocial factors on sustainable behavior in Indonesia. *Heliyon*, *10*(18).

[21] Gao, R., Zhang, H., Gong, C., & Wu, Z. (2022). The role of farmers' green values in creation of green innovative intention and green technology adoption behavior: Evidence from farmers grain green production. *Frontiers in Psychology*, *13*, 980570.

[22] Chowdhury, H., Argha, D. B. P., & Ahmed, M. A. (2023). Artificial intelligence in sustainable vertical farming. *arXiv preprint arXiv:2312.00030*.

[23] Zhang, K., Ning, Y., & Zhou, Y. AI Technology Accessibility's Impact on College Students' Academic Achievement through AI Literacy. *Yimin and Zhou, Ying, AI Technology Accessibility's Impact on College Students' Academic Achievement through AI Literacy*.

[24] Adewusi, A. O., Asuzu, O. F., Olorunsogo, T., Iwuanyanwu, C., Adaga, E., & Daraojimba, D. O. (2024). AI in precision agriculture: A review of technologies for sustainable farming practices. *World Journal of Advanced Research and Reviews*, *21*(1), 2276-2285.

[25] Omotayo, A. O., Adediran, S. A., Omotoso, A. B., Olagunju, K. O., & Omotayo, O. P. (2025). Artificial intelligence in agriculture: ethics, impact possibilities, and pathways for policy. *Computers and Electronics in Agriculture*, *239*, 110927.